The Complete Structural Mereology of a Neurosymbolic Search-and-Rescue Manifold: A Unified Axiomatic System Specification

System Architecture Invariant Manifest

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1 Introduction and Architectural Core

We formalize the search-and-rescue plot for a central Fire-Station as an interlocking multi-graph Sequential Dynamic System (SDS) operating across distinct modal boundaries. Following Edward Zalta's Theory of Abstract Objects, the spatiotemporal execution plane maps to ordinary object exemplification (Fx), while the invariant control forests encode properties atemporally (xF).

2 Decoupling of Brutal Fact Tokens and Strategic Policy Trees

To prevent semantic dilution during multi-agent transmission loops, the architecture strictly decouples raw narrative components from abstract action invariants:

1. **Literal Exemplar Tokens (Brute Facts):** The spatiotemporal particulars, file strings, and token assignments live localized within the agent's `exs.pl` file structure. This represents the extensional ground plane of the story.
2. **Strategic Reward Metrics (Intensional Policy):** The decision trees ($\mathcal{T}_k$) store only the non-linear Deep Reinforcement Learning reward coefficients ($\boldsymbol{r}_1, \boldsymbol{r}_2$), progression metrics, and action commands (π).

The tree vertices classify the *structural form* of the problem by analogy, leaving the literal variables sandboxed. Consequently, when an MCP theory migration is triggered, the system transmits an optimized strategic policy map rather than a relational data fragment, preserving out-of-distribution transferability.

3 The Eight Rasiowa Satisfaction Clauses for Multi-Agent Consensus

Let $\mathcal{M} = \langle A_t \rangle_{t \in T}$ be a T-reality over a branching poset lattice where $w \preceq t$. The consensus satisfaction relation $\mathcal{M} \models d_t(\phi)[v]$ under variable valuation v unrolls across eight rigorous algebraic operators:

1. $\mathcal{M} \models d_t(p_i(x_1, \ldots, x_m))[v] \iff A_t \models p_i(x_1, \ldots, x_m)[v]$
2. $\mathcal{M} \models d_t(\phi \vee \psi)[v] \iff \mathcal{M} \models d_t(\phi)[v]$ or $\mathcal{M} \models d_t(\psi)[v]$

3. $\mathcal{M} \models d_t(\phi \wedge \psi)[v] \iff \mathcal{M} \models d_t(\phi)[v]$ and $\mathcal{M} \models d_t(\psi)[v]$
4. $\mathcal{M} \models d_t(\phi \implies \psi)[v] \iff \forall s \preceq t, \mathcal{M} \models d_s(\phi)[v] \implies \mathcal{M} \models d_s(\psi)[v]$
5. $\mathcal{M} \models d_t(\neg\phi)[v] \iff \forall s \preceq t$, not $(\mathcal{M} \models d_s(\phi)[v])$
6. $\mathcal{M} \models d_s(d_t(\phi))[v] \iff \mathcal{M} \models d_t(\phi)[v]$
7. $\mathcal{M} \models d_t(\forall x_i \phi)[v] \iff \forall a \in \mathcal{M}, \mathcal{M} \models d_t(\phi)[v(i/a)]$
8. $\mathcal{M} \models d_t(\exists x_i \phi)[v] \iff \exists a \in \mathcal{M}, \mathcal{M} \models d_t(\phi)[v(i/a)]$